

RSA Encryption – Theory (With Example)

Introduction

RSA (Rivest–Shamir–Adleman) is a widely used asymmetric cryptographic algorithm. It uses two keys: Public Key for encryption and Private Key for decryption. The security of RSA is based on the difficulty of factorizing large prime numbers.

Key Generation Process

1. Choose two prime numbers: $p = 17$, $q = 11$
2. Compute $n = p \times q = 187$
3. Compute $\phi(n) = (p - 1)(q - 1) = 160$
4. Choose e such that $\gcd(e, \phi(n)) = 1 \rightarrow e = 7$
5. Compute d such that $d \equiv e^{-1} \pmod{\phi(n)} \rightarrow d = 23$

Keys

Public Key: $(7, 187)$

Private Key: $(23, 187)$

Encryption Process

Each character is converted into ASCII value.

$H = 72$, $I = 73$

Encryption formula: $C = M^e \pmod{n}$

Encrypt 'H': $72^7 \pmod{187} = 30$

Encrypt 'I': $73^7 \pmod{187} = 61$

Encrypted Output

"HI" $\rightarrow [30, 61]$

Decryption Process

Decryption formula: $M = C^d \pmod{n}$

Decrypt 30: $30^{23} \bmod 187 = 72 \rightarrow H$

Decrypt 61: $61^{23} \bmod 187 = 73 \rightarrow I$

Mathematical Principle

RSA works based on modular arithmetic and Euler's theorem:

$$M^{(e \times d)} \equiv M \bmod n$$

Since $e \times d = 161$ and $161 \equiv 1 \bmod 160$, we get $M^{161} \equiv M \bmod 187$

Important Notes

- 1 Message must be less than n
- 2 Small primes are used only for learning
- 3 Real RSA uses large primes and padding techniques

Conclusion

RSA is a powerful encryption algorithm that uses two keys and relies on number theory to ensure secure communication.